Unified chassis control: a model-free approach

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Abstract

Automotive safety has been a critical focus for both national and international research institutions in recent years. The combined control of lateral dynamics and roll stability in vehicles has been extensively studied, often relying on automotive models to design and tune controllers. However, these models frequently include parametric uncertainties, nonlinear dynamics, and simplifying assumptions, which can lead to suboptimal controller performance in real-world applications. This work proposes a unified chassis controller (UCC) to regulate lateral and roll dynamics during abrupt maneuvers and evaluates a model-free control (MFC) strategy against representative benchmark controllers. The comparison includes a fixed-parameter 2PID+fuzzy controller, a gain-scheduled linear quadratic regulator (LQR), and the built-in CarSim ESC baseline, all assessed within the same co-simulation environment combining MATLAB/Simulink® and CarSim®. Results demonstrate that while the benchmark controllers degrade under varying road conditions, the MFC-based UCC achieves superior robustness and performance, effectively regulating the state variables across different scenarios. These findings highlight the robustness and adaptability of MFCs, making them a promising solution for enhancing vehicle safety.

Keywords: UCC, robustness, CarSim, MFC

1 Introduction

Road traffic accidents remain a major global concern, causing approximately 1.3 million deaths and tens of millions of injuries each year (Naciones Unidas (Noticias ONU), 2022). Many of these incidents are associated with loss of vehicle stability during abrupt maneuvers, tire failures, or adverse road conditions (Friedman et al., 2010). Such situations are typically characterized by excessive side-slip angle and yaw rate, which can lead to oversteer or understeer, as well as increased rollover risk due to significant roll angle variations (Stolle et al., 2011; Zhan, 2017; Nguyen, 2023). Consequently, side-slip angle, yaw rate, and roll angle have been widely recognized as the key state

variables for vehicle stability monitoring and control, serving as the primary feedback signals in active safety systems (Rajamani, 2012).

To address these challenges, modern vehicles incorporate advanced control systems such as Electronic Stability Control (ESC), which improves lateral stability through selective braking and torque modulation (Aleksander and Edward, 2007). In parallel, suspension-based systems, including active and semi-active suspensions, are used to regulate vertical dynamics and enhance ride comfort.

Unified Chassis Control (UCC) extends these concepts by coordinating multiple subsystems to simultaneously regulate lateral and vertical vehicle dynamics, improving overall stability and safety, particularly under extreme driving conditions (Wanki et al., 2012). However, achieving this coordination is non-trivial, as lateral stability enhancement, rollover mitigation, and actuator allocation are inherently competing objectives, requiring careful trade-off management to avoid control conflicts between subsystems (Falcone et al., 2007).

Most existing UCC strategies rely on model-based control approaches, which can be broadly categorized by their underlying vehicle representation. On one end, simplified formulations such as the bicycle model have been widely adopted within Model Predictive Control (MPC) frameworks, offering computational tractability for real-time path-following and yaw stabilization tasks (Ahmadi, 2021; Gidlewski et al., 2017; Yakub et al., 2016). On the other end, multi-degree-of-freedom formulations have been proposed to capture the coupling between lateral, vertical, and suspension dynamics more faithfully: Lai and Jiang (2021) developed an 18-DOF chassis model that enables high-fidelity simulation of cross-subsystem interactions, while Chen et al. (2021) extended this perspective to electric vehicles by explicitly incorporating wheel vertical vibrations into a UCC architecture.

Although these high-fidelity representations improve prediction accuracy and allow for a more principled treatment of lateral-vertical coupling, they introduce substantial computational overhead and exhibit strong sensitivity to parametric uncertainty, limiting their portability across varying road conditions and vehicle configurations (Guo et al., 2020; Lew et al., 2024).

To mitigate these limitations, several works have introduced uncertainty compensation mechanisms based on data-driven or adaptive techniques. Neural network-based controllers have been widely used to approximate unknown nonlinearities and disturbances in vehicle dynamics, enhancing robustness under varying conditions. Similarly, fuzzy and neuro-fuzzy approaches enable online adaptation of control parameters, while sliding mode control combined with intelligent approximators provides robustness against disturbances and model mismatch (Swain et al., 2021; Guo et al., 2021).

More recently, hybrid strategies combining model-based controllers with learning-based compensation terms have demonstrated improved performance in the presence of uncertainties (Levine et al., 2016). While these strategies collectively advance robustness against parametric uncertainty and external disturbances, a systematic and homogeneous comparison between them and model-free control approaches within a UCC architecture remains scarce in the literature, limiting the ability to draw generalizable conclusions about their relative merits under consistent evaluation conditions (Sánchez-Morales et al., 2023; Mazzilli et al., 2021).

Despite the advances described above, the practical deployment of UCC strategies under varying operating conditions remains challenging. Model-based approaches, even when augmented with adaptive or learning-based compensation mechanisms, often remain sensitive to parametric uncertainty and typically require significant retuning when vehicle parameters or road conditions change.

In this context, model-free control (MFC) has emerged as a structurally simpler alternative that reduces dependence on an explicit vehicle model; however, its performance within a UCC architecture has only rarely been assessed against representative baselines. In particular, many comparative studies in this domain rely primarily on simplified fixed-parameter benchmarks, often limited to lateral PID-based ESC schemes, leaving open the question of how MFC compares with more meaningful references such as gain-scheduled controllers and, simulator-embedded

baseline strategies alternatives under consistent evaluation conditions.

To address this gap, this paper proposes a model-free UCC strategy for the simultaneous regulation of side-slip angle, yaw rate, and roll angle, and evaluates it in a MATLAB/Simulink-CarSim co-simulation environment under a sine-with-dwell maneuver with different road friction coefficients. The main contributions of this work are threefold: first, the design and implementation of an MFC-based UCC architecture for the joint regulation of lateral and vertical vehicle dynamics without relying on an explicit vehicle model; second, a comparative assessment against representative benchmark controllers, including a fixed-parameter 2PID+fuzzy controller, a gain-scheduled LQR, and a CarSim built-in baseline; and third, an analysis of practical closed-loop stability, robustness to friction variation, and control performance limits of the proposed approach.

The remainder of the paper is organized as follows. Section 4.6 presents the modeling framework and reviews model-based UCC approaches. Section 5 introduces the fundamentals of MFC and the design of intelligent PID (iPID) controllers. Section 6 describes the proposed UCC architecture. Section 7 discusses controller tuning, and Section 8 presents the experimental setup. Finally, Section 9 discusses the results, and Section 10 concludes the paper.

2 Control Problem Formulation

This section formally defines the control problem addressed in this work. The regulated variables, reference generation strategy, actuator constraints, and maneuver evaluation criteria are described, establishing the framework within which the proposed UCC is designed and assessed.

2.1 Vehicle States and Control Objectives

The vehicle dynamics relevant to this work are characterized by three state variables: the side-slip angle β, the yaw rate $\dot{\psi}$, and the roll angle ϕ. Together, these variables capture the coupled lateral and vertical behavior of the chassis during abrupt maneuvers and serve as the primary feedback signals for the proposed UCC.

The side-slip angle is defined as:

$$\beta = \tan^{-1}\left(\frac{V_y}{V_x}\right), \tag{1}$$

where V_y and V_x are the lateral and longitudinal vehicle velocities, respectively. In the present co-simulation setup, both V_x and V_y are directly obtained from CarSim. Accordingly, β is computed from ideal state information available in simulation, rather than being reconstructed from measured signals through an estimation algorithm. This choice is suitable for comparative controller assessment, although it abstracts from the state-estimation problem that would arise in a real-vehicle implementation.

The reference for β is set to zero, reflecting the control objective of minimizing lateral drift. The controller acts only when the side-slip angle exceeds the safety threshold:

$$|\beta| > \beta_{\text{lim}} = 2^\circ. \tag{2}$$

The yaw rate reference $\dot{\psi}_D$ is computed dynamically based on longitudinal speed, front wheel steering angle, and road friction coefficient, as described in Section 2.2. The roll angle reference is always set to zero, as the control objective is to minimize chassis tilt throughout the maneuver.

For the sine-with-dwell maneuver SWD considered in this study, choosing $\phi^* = 0$ is appropriate because the main control objective is to minimize body roll and preserve rollover safety margins during a severe transient maneuver, rather than to track a nonzero equilibrium roll angle. Under this perspective, a zero roll reference represents the nominal upright posture of the vehicle and allows the controller to directly counteract excessive lateral load transfer.

2.2 Reference Generation and Operating Constraints

The desired yaw rate is computed as:

$$\dot{\psi}_D = \min\left(\left|\frac{\mu g}{V_x}\right|, \left|G_{\dot{\psi}}\,\delta_f\right|\right) \cdot \text{sign}(\delta_f), \tag{3}$$

where $G_{\dot{\psi}} = V_x/(L + K_{us}V_x^2)$ is the yaw rate gain, L is the vehicle wheelbase, K_{us} is the understeer

coefficient, δ_f is the front wheel steering angle, μ is the road friction coefficient, and g is the gravitational acceleration. The parameters required to compute $G_{\dot{\psi}}$ are taken from Arronte et al. (2023). The same reference provides the full set of vehicle parameters used in the CarSim model considered in this work.

In this study, the road friction coefficient μ is imposed as a simulation parameter and kept constant within each test scenario. Therefore, μ is not estimated online, but prescribed a priori in CarSim to define the operating condition under which the controllers are evaluated. This choice is consistent with the main objective of the experiments, namely, to assess controller performance and robustness under different known friction levels.

2.3 Control Inputs and Actuator Constraints

The high-level controller produces two virtual control inputs: a yaw moment M_z, acting about the vehicle's vertical axis to regulate lateral dynamics, and a roll moment M_x, acting about the longitudinal axis to regulate vertical dynamics. These moments are subject to the following saturation constraints:

$$M_z \in [-8000,\ 8000]\ \text{Nm}, \tag{4}$$

$$M_x \in [-1500,\ 1500]\ \text{Nm}. \tag{5}$$

The saturation limits were set in accordance with the maximum torque capacity of the ABS and AARB subsystems as defined in the CarSim® SUV model used throughout this work, consistent with the simulation framework established in (Arronte et al., 2023).

At the actuation level, M_z is decomposed into individual wheel brake pressure commands by the ABS subsystem, while M_x is translated into active anti-roll bar (AARB) force commands. Both decompositions are handled internally by CarSim®, as described in Section 6.

2.4 Maneuver Completion and Failure Criteria

The performance of the UCC is evaluated under a Sine With Dwell (SWD) maneuver, a standardized test used to assess vehicle response to sudden evasive inputs, as specified in FMVSS 126 and UN/ECE Regulation No. 13-H (Mechanical Simulation Corporation, 2012). A maneuver is considered successfully completed if the vehicle maintains controlled motion throughout the full duration of the steering input without loss of stability.

In the context of the SWD maneuver, maneuver failure is determined following the criteria established by FMVSS 126 and implemented in CarSim® (Mechanical Simulation Corporation, 2021): the vehicle is considered to have failed the maneuver if the yaw rate measured at 1.0 s after the completion of steering exceeds 35% of the peak yaw rate recorded during the maneuver, or if the yaw rate at 1.75 s after the completion of steering exceeds 20% of that peak value (Mechanical Simulation Corporation, 2012). In practice, within the CarSim co-simulation environment, this condition manifests as an abrupt termination of the simulation due to vehicle loss of control, which is the failure indicator used throughout Section 9.

In addition to maneuver completion, controller performance is quantified using the root mean square (RMS) of each regulated variable, the root mean square error (RMSE) with respect to the corresponding reference, and the maximum absolute value (MAX) recorded during the maneuver. Control effort is monitored through the time histories of M_z and M_x, with particular attention to saturation occurrence and duration.

3 Benchmark Controllers and Comparative Framework

To provide a rigorous and meaningful assessment of the proposed MFC-based UCC, three benchmark controllers are considered. Each controller operates within the same UCC architecture described in Section 6, receives the same input signals, and is subject to the same actuator saturation constraints defined in Section 2.3. The benchmarks are selected to represent progressively more capable reference strategies, allowing the comparison to go beyond the fixed-gain baselines that are

common in the literature (Sánchez-Morales et al., 2023).

3.1 Fixed-Parameter 2PID+Fuzzy Controller

The first benchmark is the fixed-parameter UCC architecture reported in (Arronte Delgado, 2024). In that design, the lateral dynamics are regulated by two independent PID controllers with hysteresis acting on the side-slip angle β and the yaw rate $\dot{\psi}$. Their outputs are combined into a corrective yaw moment M_z, which is implemented by differential braking in the ESC layer. As in the earlier ESC design, the lateral PID branches are only activated when the corresponding regulated variables leave prescribed admissible bands, which reduces unnecessary braking effort during mild transients.

The reference described in (Arronte Delgado, 2024) forms an integrated "2PID+fuzzy" structure. The vertical dynamics are handled by a Mamdani-type fuzzy controller that uses the roll angle ϕ and the roll rate $\dot{\phi}$ as inputs and generates the corrective roll moment M_x as output. In addition, the fuzzy layer is complemented by a feedforward stage intended to attenuate the effect of lateral-force disturbances on the roll dynamics.

Accordingly, this benchmark acts on the same two control moments considered in the present UCC framework, namely M_z for lateral stabilization and M_x for roll mitigation. It therefore provides a fixed-parameter reference for the joint lateral-roll regulation problem addressed in this paper, while retaining a comparatively simple and interpretable control structure.

3.2 Gain-Scheduled LQR Controller

The second benchmark is a gain-scheduled linear quadratic regulator (LQR), in which the state-feedback gain matrix is adapted as a function of the longitudinal vehicle speed V_x.

The LQR benchmark is derived from the bicycle-model framework described in (Arronte et al., 2023), but extended to represent the coupled lateral and roll dynamics required by the present UCC problem. In particular, the state-space model is written in the descriptor form $E\dot{x} + Fx = Gu$, with state vector $x = [v\ \dot{\psi}\ \dot{\phi}\ \phi]^\top$ and input vector $u = [\delta\ M_z\ M_x]^\top$, where δ is the steering input, M_z is the external yaw moment, and M_x is the external roll moment. Relative to the model used in (Arronte et al., 2023), the main modification lies in the input matrix G, which is reformulated to account simultaneously for steering excitation and for the two control moments used by the UCC architecture. This yields a speed-dependent linear model that preserves the bicycle-model interpretation while explicitly capturing the actuation channels required for both lateral stabilization and roll regulation.

The scheduling structure is defined over 11 longitudinal-speed operating points, from 20 km/h to 120 km/h in increments of 10 km/h. For each point, a state-feedback gain matrix is precomputed, and the gains used during simulation are obtained by interpolating these matrices in MATLAB using the `interp1` function as a function of the measured vehicle speed V_x. The resulting gain-scheduling mechanism is applied to the complete controlled system, so that both the lateral dynamics and the roll dynamics are governed by speed-dependent state feedback. The longitudinal speed V_x is selected as the scheduling variable because, in the modified bicycle model used to derive the LQR, the system matrices depend explicitly on V_x, which directly changes the lateral tire-force terms, yaw dynamics, and the overall lateral-roll coupling.

The weighting matrices were selected following a normalized design logic similar to Bryson's rule, in which each state and input is penalized according to an admissible maximum value. For the state vector $x = [v\ \dot{\psi}\ \dot{\phi}\ \phi]^\top$, the adopted limits were $v_{\max} = 0.5\,\mathrm{m/s}$, $\dot{\psi}_{\max} = 0.3\,\mathrm{rad/s}$, $\dot{\phi}_{\max} = 0.2\,\mathrm{rad/s}$, and $\phi_{\max} = 0.08\,\mathrm{rad}$, leading to $Q = \mathrm{diag}(1/v_{\max}^2,\ 1/\dot{\psi}_{\max}^2,\ 1/\dot{\phi}_{\max}^2,\ 1/\phi_{\max}^2)$. For the input vector $u = [\delta\ M_z\ M_x]^\top$, the admissible values were taken as $\delta_{\max} = 0.1\,\mathrm{rad}$, $M_{z,\max} = 8000\,\mathrm{Nm}$, and $M_{x,\max} = 1500\,\mathrm{Nm}$, so that $R = 0.1\,\mathrm{diag}(1/\delta_{\max}^2,\ 1/M_{z,\max}^2,\ 1/M_{x,\max}^2)$. The scalar factor 0.1 was introduced as a global aggressiveness coefficient to allow a more responsive controller without over-penalizing the control action. For each scheduling point, the descriptor model is converted to the standard form $\dot{x} = A_{\mathrm{sys}}x + B_{\mathrm{sys}}u$ with $A_{\mathrm{sys}} = -E^{-1}F$ and $B_{\mathrm{sys}} = E^{-1}G$, and the corresponding feedback gain is then obtained from $K_{\mathrm{LQR}} = \mathrm{lqr}(A_{\mathrm{sys}}, B_{\mathrm{sys}}, Q, R)$.

This controller represents a more capable baseline than the fixed-parameter 2PID+fuzzy controller, as it exploits a linearized vehicle model and schedules the state-feedback gains as a function of the longitudinal speed V_x. Its inclusion allows the comparison to contrast the MFC's online disturbance estimation against a structured model-based benchmark with speed-dependent state feedback.

3.3 CarSim Built-in Baseline Controller

The third benchmark is the ESC baseline controller embedded in CarSim®, which provides a representative Electronic Stability Control example for FMVSS 126 / ECE R13H sine-with-dwell testing (Mechanical Simulation Corporation, 2025).

As described in the official CarSim FMVSS 126 application example, this built-in baseline is intended for ESC evaluation under the sine-with-dwell procedure and is assessed through lateral-stability criteria based mainly on yaw-rate decay after the steering input, together with the prescribed lateral-displacement check for high-gain cases (Mechanical Simulation Corporation, 2025). The public description does not indicate any dedicated roll-control channel; therefore, this baseline is interpreted here as an ESC-oriented lateral-stability controller rather than as a full unified chassis controller with explicit roll regulation.

The public CarSim description documents the test procedure and the example workflow, but it does not disclose the internal implementation of the embedded ESC in sufficient detail to establish whether the controller uses additional simulator-internal variables beyond the documented test signals (Mechanical Simulation Corporation, 2025). For that reason, the CarSim baseline should be viewed as a simulator-native black-box reference. This makes it valuable as a practical benchmark, but also means that its informational symmetry with the other controllers cannot be fully verified from the publicly available documentation.

This baseline is particularly relevant because it represents a industry-relevant reference implemented within the same simulation environment, providing a practical indication of the performance margin achievable with a conventional ESC-oriented strategy under identical test conditions.

4 Model-Free Unified Chassis Control Design

This section presents the proposed MFC-based UCC architecture. The formulation begins with the ultra-local model and the intelligent PID control law, followed by the algebraic estimators for the lumped disturbance term, the overall UCC architecture, and the coordination and control allocation strategies. The section concludes with the parameter selection procedure.

4.1 Ultra-Local Model and Intelligent PID Formulation

The MFC framework adopted in this work is based on the ultra-local model introduced by Fliess and Join (2008), which assumes that the plant input-output behavior can be approximated over a short time window as:

$$y^{(v)} = F + \alpha u, \tag{6}$$

where $y^{(v)}$ denotes the v-th time derivative of the output, F is a lumped term that captures the unknown plant dynamics, external disturbances, and unmodeled nonlinearities, $\alpha \in \mathbb{R}$ is a non-physical scaling parameter chosen by the designer, and u is the control input (Fliess, 2009). The term F is treated as a piecewise-constant function updated at each estimation window, making the representation independent of any explicit vehicle model.

In this work, three plant variables are regulated: the yaw rate $\dot{\psi}$, the side-slip angle β, and the roll angle ϕ. The yaw rate channel is represented by a first-order ultra-local model ($v = 1$), while the side-slip angle and roll angle channels use a second-order formulation ($v = 2$). This choice is consistent with the order of the corresponding vehicle equations: $\dot{\psi}$ is taken as the derivative of the yaw motion, whose dynamics stem from a second-order equation and are therefore adequately captured with $v = 1$, whereas β and ϕ are governed by first-order equations, making a second-order ultra-local description more suitable for their regulation.

Assuming that the ultra-local model holds over a short time window, the stabilizing intelligent PID (iPID) control law is given by:

$$u = \frac{-F + \ddot{y}^*}{\alpha} + K_P e + K_I \int e\, dt + K_D \frac{de}{dt}, \quad (7)$$

where y^* is the reference trajectory, $e = y - y^*$ is the tracking error, and K_P, K_I, K_D are the PID gains. In this work, the integral term is set to zero ($K_I = 0$), reducing the controller to an intelligent PD (iPD) structure. The term $-F/\alpha$ acts as an online feedforward compensation that cancels the estimated lumped dynamics, while the PD component handles residual tracking errors. This choice is appropriate for the SWD maneuver because the control task is dominated by transient stabilization rather than by the elimination of steady-state error.

4.2 Algebraic Estimation of the Lumped Term

The real-time estimation of F is performed using an algebraic approach derived from the ultra-local model via Laplace-domain manipulations, following Fliess (2009). The estimators are expressed as weighted integrals of the input and output signals over a sliding window of length T, which provides inherent noise attenuation without requiring explicit filtering.

4.2.1 Second-order estimator for β and ϕ

For the second-order channels ($v = 2$), the lumped term is estimated as:

$$F_2 = \frac{60}{t^5} \int_0^t (t^2 - 6t\tau + 6\tau^2) y(\tau)\, d\tau - \frac{30\alpha}{t^5} \int_0^t (t-\tau)^2 \tau^2 u(\tau)\, d\tau. \quad (8)$$

4.2.2 First-order estimator for $\dot{\psi}$

For the first-order channel ($v = 1$), the estimator takes the form:

$$F_1 = -\frac{6}{t^3} \int_0^t (t - 2\tau) y(\tau)\, d\tau - \frac{6\alpha}{t^3} \int_0^t (t-\tau)\tau\, u(\tau)\, d\tau. \quad (9)$$

4.2.3 Discrete-time implementation

For digital implementation, the integrals in Eqs. (17) and (20) are approximated using the trapezoidal rule over a window of length T, sampled at period T_s. The discrete estimators are:

$$\hat{F}_2 = \frac{60}{T^5} \sum_{k=0}^{T/T_s} a_k (T^2 - 6TkT_s + 6(kT_s)^2) y[k] - \frac{30\hat{\alpha}}{T^5} \sum_{k=0}^{T/T_s} a_k (T - kT_s)^2 (kT_s)^2 u[k], \quad (10)$$

$$\hat{F}_1 = -\frac{6}{T^3} \sum_{k=0}^{T/T_s} a_k (T - 2kT_s) y[k] - \frac{6\hat{\alpha}}{T^3} \sum_{k=0}^{T/T_s} a_k (T - kT_s) kT_s\, u[k], \quad (11)$$

where the trapezoidal weights are:

$$a_k = \begin{cases} T_s/2, & k = 0 \text{ and } k = T/T_s, \\ T_s, & k = 1, \ldots, T/T_s - 1. \end{cases} \quad (12)$$

The integration window was set to $T = 1.0\,\mathrm{s}$ for all three estimators, based on the characteristic response time of the vehicle dynamics under the SWD maneuver. This value represents a compromise between estimation smoothness and responsiveness: longer windows improve the accuracy and noise attenuation of $\hat{F}$ but slow down the estimator, whereas shorter windows yield faster convergence at the expense of greater sensitivity to measurement noise; in the present SWD conditions, $T = 1.0\,\mathrm{s}$ provided adequate filtering without introducing excessive delay in the transient control action.

4.3 Unified Chassis Control Architecture

An upstream Multi-layer Coordination (MuC) architecture is adopted for the UCC, as illustrated in Fig. 4. The architecture comprises four layers: supervision, high-level control, coordination, and

control allocation. The inputs to the system — longitudinal speed V_x, front wheel steering angle δ_f, lateral speed V_y, yaw rate $\dot{\psi}$, roll angle ϕ, and roll rate $\dot{\phi}$ — are obtained directly from CarSim®.

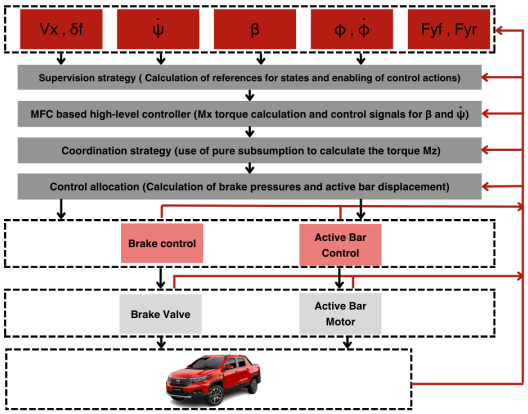


Fig. 1: UCC architecture

4.3.1 Supervision layer

The supervision layer computes the references and defines the activation conditions for each controller channel, as formalized in Section 2.2. The β controller is activated only when $|\beta| > 2°$; the $\dot{\psi}$ reference is computed dynamically via Eq. (3); and the ϕ reference is always zero.

4.3.2 High-level controller

The high-level controller applies three independent iPD controllers — one per regulated variable — to compute the virtual control moments. The lateral dynamics channels (β and $\dot{\psi}$) produce a combined yaw moment M_z, while the roll channel produces a roll moment M_x. Both are subject to the saturation limits defined in Eqs. (4)–(5). The implementation in MATLAB/Simulink® is shown in Fig. 5.

4.3.3 Coordination layer

Since both the β and $\dot{\psi}$ controllers contribute to M_z, a Pure Subsumption strategy is used in the coordination layer to weight their respective contributions in the final torque signal. The roll channel is independent and not affected by this coordination step.

4.4 Control Allocation

The virtual control moments M_z and M_x are translated into physical actuator commands by CarSim®. Specifically, M_z is decomposed into individual wheel brake pressure signals processed by the ABS subsystem, and M_x is converted into active anti-roll bar force commands. This allocation step is handled internally by the simulator, consistent with the setup described in (Arronte et al., 2023).

4.5 Parameter Selection and Implementation Details

The iPD controllers involve three sets of parameters per channel: the scaling constant α, the proportional gain K_P, and the derivative gain K_D. These were selected empirically through iterative simulation, guided by the following criteria: α was adjusted to match the order of magnitude of the unknown plant gain for each channel, while K_P and K_D were tuned to balance tracking performance and control signal smoothness, prioritizing maneuver safety over passenger comfort.

The effect of α on closed-loop behavior was observed to be monotonic: higher values introduce response delay, while lower values increase control signal oscillation. An intermediate value was sought for each channel independently. It was further verified through simulation that variations in K_D had negligible influence on the system response; therefore, K_D was fixed at a small value for all channels to provide light damping.

The resulting parameter values are summarized in Table **??**.

The simulation sampling period was set to $T_s = 1\,\text{ms}$, and the estimation window to $T = 1.0\,\text{s}$ for all three channels.

4.6 UCC model-based

Currently, most of the UCC systems use vehicle models to obtain and adjust the parameters of the individual controllers.

A widely adopted approach is the bicycle model, illustrated in Fig. 3. In this representation, F_f and F_r denote the lateral forces acting on the front and rear axles, respectively; V_{ref} is the vehicle's velocity vector; and α_f and α_r are the slip angles of the front and rear wheels. The Center of Gravity (CG) is located at distances l_f and l_r from

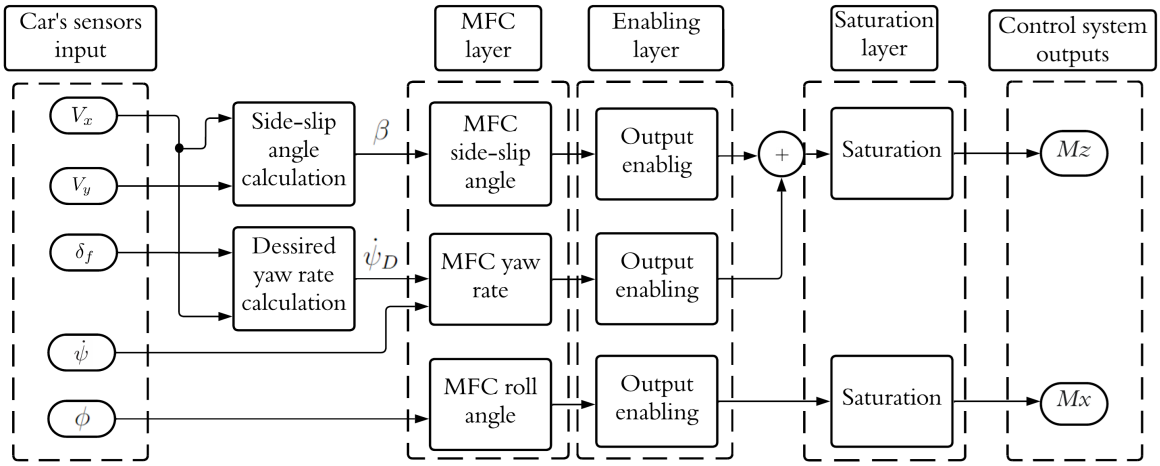


Fig. 2: UCC implementation

the front and rear axles. The side-slip angle, β, represents the relationship between the vehicle's longitudinal and lateral velocities, while $r = \dot{\psi}$ denotes the yaw rate. Additionally, δ_f epresents the front wheel steering angle, and V_f and V_r are the velocity vectors at the front and rear tires.

This model is frequently integrated into Model Predictive Control (MPC) frameworks, as seen in Ahmadi (2021), Gidlewski et al. (2017), and Yakub et al. (2016). Other studies, such as Owen (2018), utilize it to analyze the impact of the center-of-mass location on vehicle performance, while Bilen (2010) and Fahimi (2013) employ it for computational simulations and controller design (Xia et al., 2021).

To achieve higher fidelity, some authors develop more complex models to capture intricate vehicle dynamics and subsystem interactions. For instance, Lai and Jiang (2021) introduced an 18-degree-of-freedom (DOF) model to represent comprehensive chassis dynamics. This model facilitates the design of a UCC system that evaluates the trade-off between ride comfort and handling stability across various maneuvers, accounting for the complex coupling between subsystems. The 18-DOF model treats the vehicle as five rigid bodies: a sprung mass (chassis) with 6-DOF (roll, pitch, yaw, and translational motions) and four unsprung masses (wheels). Each wheel is modeled with 3-DOF, corresponding to vertical displacement, wheel rotation, and steering rotation.

Similarly, Chen et al. (2021) proposed a 6-DOF rigid-body model that integrates planar motion (longitudinal, lateral, and yaw) with spatial motion (vertical, roll, and pitch). Combined with unsprung mass dynamics, this approach treats the vehicle as a unified system under longitudinal and lateral accelerations. To characterize nonlinear tire behavior, the authors utilize Pacejka's Magic Formula, an empirical approach that unifies longitudinal force, lateral force, and self-aligning moment through parameter sets fitted to experimental data.

The use of model-based controls in vehicle dynamics offers the fundamental advantage of enabling precise and coordinated management of complex multivariable systems—such as the interaction between independent wheel drive and active suspension—facilitating the incorporation of physical constraints and the prediction of critical states to maximize safety and comfort Li et al. (2022). However, this approach has critical drawbacks, primarily a high demand for computational resources to process high-fidelity models in real time Guo et al. (2020), as well as significant sensitivity to modeling errors (model mismatch) Lew et al. (2024). Any discrepancy between mathematical parameters and real-world conditions—such as variations in mass or unexpected changes in tire friction—can significantly degrade system performance and stability if robust adaptation mechanisms are not in place.

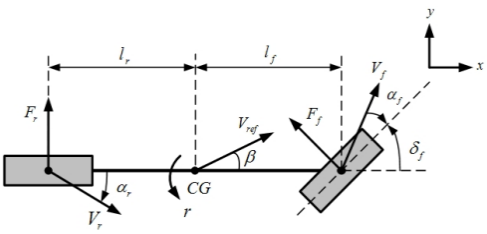


Fig. 3: Bicycle model

4.7 Uncertainty Compensation

Given the limitations discussed above, neural network-based controllers have been widely used to compensate for model uncertainties and unmodeled nonlinearities in vehicle dynamics. This is the case in Swain et al. (2021), where the lateral dynamics are described using a dynamic bicycle model, capturing key variables such as longitudinal velocity, slip angle, yaw rate, and tire lateral forces, while also accounting for trajectory tracking errors and unmodeled external disturbances (e.g., crosswinds). To compensate for parametric uncertainties, nonlinearities, and unmodeled dynamics, a radial basis function neural network (RBFNN) is integrated into a higher-order sliding mode control (HOSMC) scheme. The RBFNN estimates the unknown dynamics online based on the sliding surface, with its parameters updated through Lyapunov-based adaptive laws. This combined approach enhances robustness against disturbances, reduces chattering, and preserves lateral control performance under adverse conditions.

In addition to neural network-based approaches, fuzzy logic has also been employed to handle parametric uncertainties within sliding mode control frameworks. This is the case in Guo et al. (2021), where a fuzzy inference system is used to adaptively tune the boundary layer thickness based on the magnitude of the sliding surface, as part of a neuro-fuzzy adaptive sliding mode control strategy.

In this approach, the absolute value of the sliding surface is used as the input to the fuzzy system, while the boundary layer thickness is the output. Using predefined linguistic rules and membership functions, the controller adjusts this parameter online: large deviations from the sliding surface lead to a thinner boundary layer to ensure fast convergence, whereas small deviations result in a thicker layer to attenuate chattering.

This adaptive fuzzy mechanism enables an effective trade-off between robustness and oscillation reduction, improving control performance in the presence of parametric uncertainties and external disturbances.

One of the most widely used schemes for validating controller design in automotive systems is the CarSim® software. CarSim® mathematical models are highly optimized for computational efficiency and are represented at the system level, meaning that vehicle data is intended to be measured or calculated and does not rely on detailed knowledge of component materials or suspension linkage complexities. When using a CarSim® model, major issues may include the need for detailed vehicle data, accurate environmental conditions, and a good understanding of vehicle models at the system level. Users should also be aware of the importance of accurate kinematics and suspension compliance, tire strength and moment properties, and environmental conditions, as well as the potential impact of limited reference data on the accuracy of simulation results (Mechanical Simulation Corporation, 2021).

Due to the above, the study of MFC for the regulation of vehicle dynamics in real-time is of interest. It should be noted that the term "Model-free control" can be seen in the literature as referring to a wide range of controllers that can go from "classical" proportional-integral-derivative (PID) to robust and adaptive control techniques derived, for example, from neural networks, fuzzy systems, and soft computing. The type of MFC used in this paper (intelligent PID based on ultra-local model) was presented by Fliess and Join (2008), which makes it a novel technique and alternative to classical control in the automotive domain. Several authors have used this control technique as part of an automotive UCC system. In general, the MFC can be used in this field to regulate the longitudinal dynamics (direction and/or speed) of the vehicle (Dong et al., 2014), as a controller of the lateral dynamics (yaw rate and/or side slip angle) as in the case of Radac and Precup (2018), for the control of the joint lateral and longitudinal dynamics like in Menhour et al. (2018) and

Rizk et al. (2023), or for the control of the vertical dynamics, that is roll, roll rate and/or pitch (Mustafa et al., 2019).

However, there are no references in the literature to works that use this control method to jointly regulate the lateral and vertical dynamics at the same time and in a single UCC system. In the present work, an automotive UCC system is designed using only MFC controllers to minimize the side-slip angle, reducing the yaw rate error and maintaining the vertical stability corresponding to the car roll angle and roll rate. The proposed system is tuned and validated through a Sine With Dwell maneuver in CarSim® software. Those are original contributions, to the best of the authors' knowledge.

5 The ultra-local model-based MFC controller

The Model-Free Controller (MFC) based ultra-local model assumes that the system response will behave as shown in Eq. (13) for a small time window:

$$y^v = F + \alpha u, \tag{13}$$

where y^v represents the vth derivative of the output (usually $v = 1$ or $v = 2$) (Fliess, 2009), and F is the input/output ratio of a simplified and little-known model of the plant. This parameter is a piecewise constant function that is updated in each time window and also includes measured and unmeasured perturbations of the plant. On the other hand, $\alpha \in \mathbb{R}$ is a number that represents a non-physical constant parameter. This parameter is defined by the designer in such a way as to have the same order of magnitude as the unknown real parameter, and then an adjustment is made based on the system's time response. In this paper, there are three plant variables: yaw rate ($\dot{\psi}$), side-slip angle (β), and roll angle (ϕ), where the last two ones are controlled using a second-order ultra-local model, while the first one is represented by a first-order ultra-local model.

Assuming that the model represents the plant dynamics in a short time window, then the stabilizing intelligent PID control (iPID) can be computed by:

$$u = \frac{-F}{\alpha} + \frac{\ddot{y}^*}{\alpha} + K_P e + K_I \int e\, dt + K_D \frac{de}{dt}, \tag{14}$$

where $\ddot{y}^*$ is the reference second time-derivative that the output is intended to follow, $e = y - y^*$ is the system tracking error and K_P, K_I and K_D are the gains of a conventional PID controller. These gains are adjusted to cancel out the measured output and impose the desired dynamics. It should be noted that this control can be in the form of a PI or a PD as appropriate. Finally, the control method involves estimating the value of F in constant time windows. It can be demonstrated that even if the initial guess for α is incorrect, the estimator will correctly estimate the compensation of F (Pereira das Neves and Augusto Angélico, 2022). Therefore, α will only slightly affect the performance but will ensure the system stability.

5.1 F estimator to the ultra-local model of β and ϕ

As mentioned earlier, the ultra-local models of β and ϕ are considered of second order ($v = 2$). In that case, the equation would be $\ddot{y} = F_2 + \alpha u$. Applying the Laplace transform in this equation results in $s^2Y(s) - sy(0) - \dot{y}(0) = F_2/s + \alpha U(s)$. Then, by taking the derivative of the equation concerning s twice, one obtains:

$$2Y(s) + 4s\frac{d}{ds}Y(s) + s^2\frac{d^2}{ds^2}Y(s) = \\ = 2s^{-3}F_2 + \alpha\frac{d^2}{ds^2}U(s). \tag{15}$$

Finally, multiplying Eq. (15) by $\frac{1}{s^3}$ results in[1]:

$$2\frac{1}{s^3}Y(s) + 4\frac{1}{s^2}\frac{d}{ds}Y(s) + \frac{1}{s}\frac{d^2}{ds^2}Y(s) = \\ = 2F_2\frac{1}{s^6} + \alpha\frac{1}{s^3}\frac{d^2}{ds^2}U(s). \tag{16}$$

Returning to the time domain using the Laplace anti-transform, it is obtained:

[1] This process ensures that integral terms appear in every factor, thereby aiding in noise reduction.

$$F_2 = \frac{60}{t^5}\int_0^t (t^2 - 6t\tau + 6\tau^2)y(\tau)\,d\tau + \\ - \frac{30\alpha}{t^5}\int_0^t (t-\tau)^2\tau^2 u(\tau)\,d\tau. \quad (17)$$

5.2 F estimator to the ultra-local model of $\dot{\psi}$

The ultra-local model of $\dot{\psi}$ is considered of first order ($v = 1$). In that case, the equation would look like $\dot{y} = F_1 + \alpha u$. The demonstration is very similar to the estimator of second order. Applying the Laplace transform in this equation results in $sY(s) - y(0) = F_1/s + \alpha U(s)$. Then, by taking the derivative of the equation by s, one has:

$$Y(s) + s\frac{d}{ds}Y(s) = -s^{-2}F_1 + \alpha\frac{d}{ds}U(s). \quad (18)$$

Finally, multiplying the Eq. (18) by $\frac{1}{s^2}$ results in:

$$\frac{Y(s)}{s^2} + \frac{1}{s}\frac{d}{ds}Y(s) = -F_1 + \frac{\alpha}{s^2}\frac{d}{ds}U(s). \quad (19)$$

Upon returning to the time domain via the Laplace inverse transform, one obtains:

$$F_1 = -\frac{6}{t^3}\int_0^t (t - 2t\tau)y(\tau)\,d\tau + \\ - \frac{6\alpha}{t^3}\int_0^t (t-\tau)\tau u(\tau)\,d\tau. \quad (20)$$

5.3 Discretization of algebraic estimators

The computational application of the estimator is possible through the window integral approximation. For this, the size of the integration window (T) must be decided, where, in general, the larger the window, the more precise the estimated value will be. On the other hand, the control will be slower since the estimator will take longer to converge to the value. Considering the response speed of the car dynamics, the windows for all estimators were defined as $T = 1.0$ s.

Considering the trapezoidal approximation of the integral, the estimators (of first and second order) can be written as:

$$\hat{F}_2 = \frac{60}{T^5}\sum_{k=0}^{T/T_s} a_k(T^2 - 6TkT_s + 6(kT_s)^2)y[k] + \\ - \frac{30\hat{\alpha}}{T^5}\sum_{k=0}^{T/T_s} a_k(T - kT_s)^2(kT_s)^2 u[k], \quad (21)$$

and

$$\hat{F}_1 = -\frac{6}{T^3}\sum_{k=0}^{T/T_s} a_k(T - 2TkT_s)y[k] + \\ - \frac{6\hat{\alpha}}{T^3}\sum_{k=0}^{T/T_s} a_k(T - kT_s)kT_s u[k], \quad (22)$$

where:

$$a_k = \begin{cases} T_2/2, & k = 0 \text{ and } k = T/T_s \\ T_s, & k = 1, \ldots, T/T_s - 1 \end{cases}. \quad (23)$$

6 UCC structure

In this work, an upstream architecture with multi-layer Coordination (MuC) is proposed for vehicle control. Fig. 4 shows the architecture with its components. In this architecture, inputs such as the longitudinal speed, steering angle of the front wheels and the system states (lateral speed V_y, yaw rate $\dot{\psi}$, roll angle ϕ and roll rate $\dot{\phi}$) are obtained from CarSim® as it were from a real car.

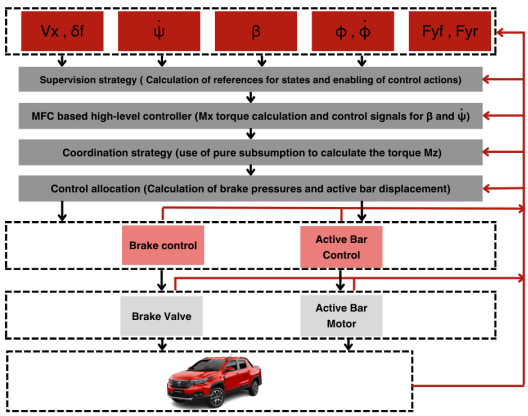


Fig. 4: UCC architecture

6.1 Supervision layer

The supervision layer aims to compute the references, as well as conditions under which the controller will act. The reference for β is zero and it is calculated based on the estimated car lateral speed, as shown in Eq. (24). It is also defined that the controller for β only performs whether it is outside the limits in Eq. (25). In the case of the yaw rate, the controller's reference is calculated based in the longitudinal speed, the steering angle of the front wheels, and the coefficient of ground friction. The controller's operating range is determined by the limits described in Eq. (26). In the case of the roll angle, the reference will always be zero, as one of the controller's objectives is to reduce this parameter. The equations are:

$$\beta = \tan^{-1}(\frac{V_y}{V_x}), \tag{24}$$

$$\beta \leq \beta_{lim} = \pm 2^\circ, \tag{25}$$

$$\dot{\psi}_D = \min\left(\left|\frac{\mu g}{V_x}\right|, \left|G_{\dot{\psi}}\delta_f\right|\right) \cdot \text{sign}(\delta_f), \tag{26}$$

where, $G_{\dot{\psi}} = V_x/(L + K_{us}V_x^2)$, V_x is the car longitudinal speed, L is the distance between the car's axles in meters and K_{us} is the under-steer coefficient.

6.2 High level controller

The high-level controller design uses the references calculated by the supervision layer and the estimated states and calculates the torques needed to control the dynamics. In the case of lateral dynamics (which involves $\dot{\psi}$ and β), the control signal will be a torque around the car's z-axis (M_z). This signal is setting a saturation level of $\pm$8000Nm. To control the roll angle, torque is calculated around the x-axis (M_x). This signal is limited by saturation levels of $\pm$1500Nm. Fig. 5 shows the implementation of this controller in Matlab/Simulink®.

6.3 Coordination layer

As can be seen in Fig. 5, in the coordination phase, a Pure Subsumption strategy is used to weigh up which of the control signals corresponding to β and $\dot{\psi}$ will have more weight in the final torque calculation. This layer is not involved in controlling the roll angle.

6.4 Control allocation

At this step, the control signals calculated in the previous steps are transformed into "understandable" variables by the subsystems that act directly on the car's chassis. In the case of the torque on the z-axis, it will be transformed into individual brake pressure signals for each wheel by the ABS. In the case of M_x, it will be transformed into suspension forces relating to the active stabilizer bar (AARB) system. This step is carried out directly by the CarSim® software.

6.5 Individual actuators

The individual brake pressure signals and the motor torque signal of the AARB system from the previous step are taken in this step as references for the controllers of the car's ABS and the active stabilizer bar system, respectively. The control of these systems finally makes it possible to establish the desired behavior of the controlled dynamics. Like the previous one, this stage is implemented by CarSim® software.

7 Control tuning

In the following subsections, a demonstration of how to adjust the parameters α, K_P, and K_D of the iPD proposed controllers is performed (in this paper, we consider the integrative factor equals to zero). It should be noted that this tuning is empirically done and that the dynamics to be controlled do not require high precision in tracking the corresponding references for the system to exhibit proper behavior.

7.1 α adjustment

This parameter is set individually for each control law, and has direct influence on the system outputs (M_z and M_x torques). Since the M_z signal is composed of the sum of individual signals corresponding to the side-slip angle control and the yaw rate control, it is necessary to adjust simultaneously α_β and $\alpha_{\dot{\psi}}$. A bad choice of these parameters can result in an oscillatory control signal with a high saturation time and/or an unsatisfactory

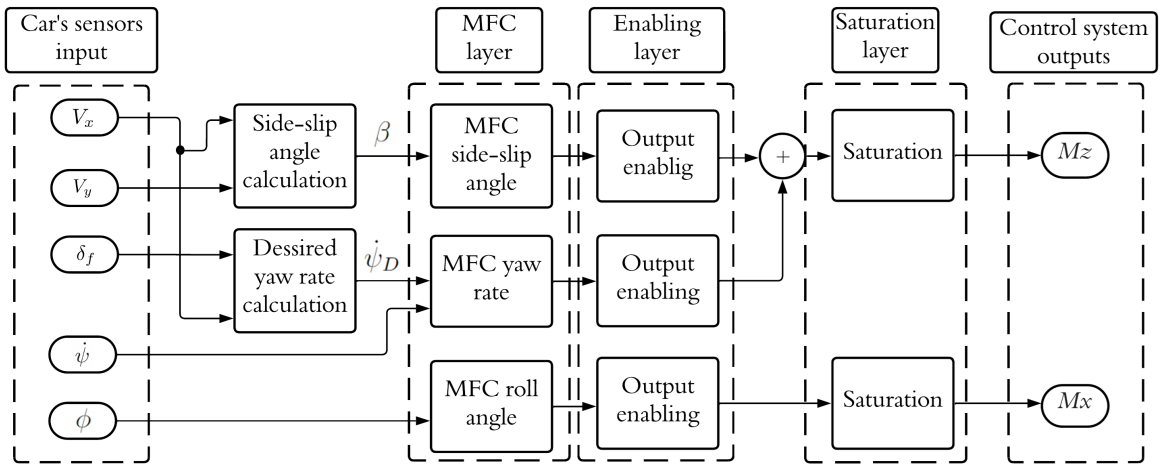


Fig. 5: UCC implementation

maneuver performance. In this work, the values of α_s are empirically obtained looking for a relationship between them and the time response of the controlled parameters. Higher α value can induce some delay in the system response while a lower α generates a more oscillatory control signal. The objective then is to find a equilibrium α. The selected α values are $\alpha_\beta = 0.01$, $\alpha_{\dot{\psi}} = 10^{-3}$ and $\alpha_\phi = 10^{-2}$.

7.2 iPD adjustment

It was verified by simulations that the variation of the K_d parameter does not have influence on the system response. The K_p was adjusted firstly for the yaw rate and side-slip angle controllers and then for the roll control. The adjustment must keep a balance between reducing the steady state error and producing an oscillatory control signal. It should be noted that in the event of a sudden maneuver (as in the case of SWD) the control must prioritize the safety over the comfort. This is evidenced by the increase in the roll rate signal. Finally, the iPD adjustment is as follows: $Kp_\beta = 20$; $Kd_\beta = 3$; $Kp_{\dot{\psi}} = 50$; $Kd_{\dot{\psi}} = 3$; $Kp_\phi = 200$; $Kd_\phi = 3$.

8 Experiment design

To validate the designed UCC, it is simulated in Matlab/Simulink® acting on the Carsim® SUV model adapted to the parameters referred in Arronte et al. (2023) in a Sine With Dwell (SWD) maneuver. The results presented in this section focus on the comparison between the proposed iPD-based MFC controller, the gain-scheduled LQR benchmark, and the open-loop vehicle response under the same test conditions. To show the system's robustness, tests are performed three different μ values (Ahrenhold et al., 2023): $\mu = 0.7$ (Dry Asphalt Road), which simulates good grip and stability for vehicles, $\mu = 0.5$ (Wet Road), which corresponds to reduced traction due to a lubricating film of water, increasing the risk of skidding, and $\mu = 0.3$ (Icy Road), which simulates extremely low traction, and requires careful driving and often needs specialized tires or chains for safety.

The SWD maneuver involves accelerating the vehicle to a speed of just over 80 km/h without steering or braking, leaving it in the highest gear up to 80 km/h, and then using a robot to apply a steering control in the form of a sine wave with a dwell (momentary pause in the maximum steering angle). The purpose of the test is to evaluate the vehicle's response to a sudden evasive maneuver, such as swerving to avoid an obstacle, and to ensure compliance with regulations such as USA FMVSS 126 and UN/ECE Regulation No. 13-H Mechanical Simulation Corporation (2012). The system simulation is performed by using CarSim® and Matlab® together. Figure 6 shows the waveform of the steering signal in this maneuver.

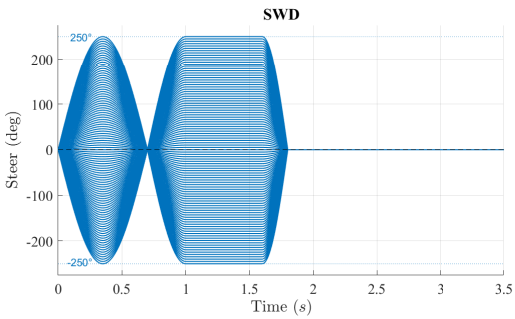


Fig. 6: Steering Angle for the Sine With Dwell maneuver

9 Simulation results

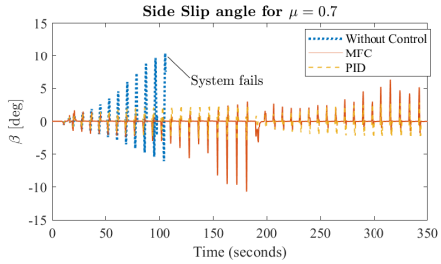
The UCC is tested by simulating a SWD maneuver as specified above for three different values of friction coefficient as described. The results below compare the open-loop situation (without controller), the gain-scheduled LQR benchmark, and the iPD MFC controller. It is shown that the first and second cases do not guarantee stability in all friction scenarios.

9.1 Simulation for $\mu = 0.7$

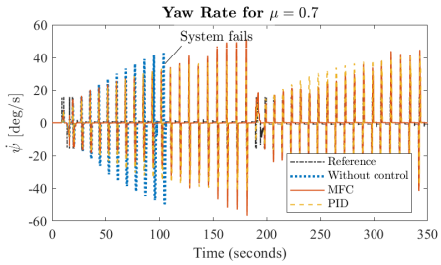
Figures 7a and 7b show the side slip angle and yaw rate in the SWD maneuver for $\mu = 0.7$. Here, both controllers, namely the gain-scheduled LQR and the iPD-based MFC, manage to decrease the side slip angle during the maneuver. Also, the measured yaw rate follows the reference for both controllers. It can be observed that the vehicle fails to complete the maneuver without the action of any controller (open-loop). Figures 7c and 7d show the roll angle and roll rate behavior during the maneuver. Both controllers manage to reduce these variables during the maneuver, thus demonstrating their effectiveness in controlling the rollover dynamics.

9.2 Simulation for $\mu = 0.5$

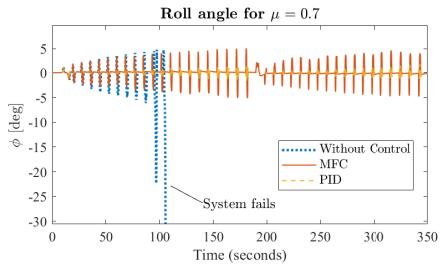
Figures 8a and 8b show the side slip angle and yaw rate respectively in the SWD maneuver with $\mu = 0.5$. Both controllers (gain-scheduled LQR and iPD MFC) effectively reduce the side-slip angle and ensure that the yaw rate follows its reference. As in the previous simulation, it can be observed that the car does not manage to complete the maneuver in open-loop (sudden stops in the curves). Figures 8c and 8d show the behavior of the roll angle and roll rate respectively. Here it is also evident the positive influence of the designed controllers in the reduction of the key parameters in the car rollover dynamics.



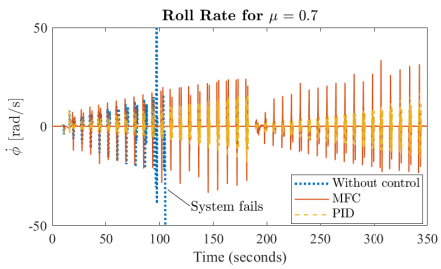
(a) Side-slip angle



(b) Yaw rate

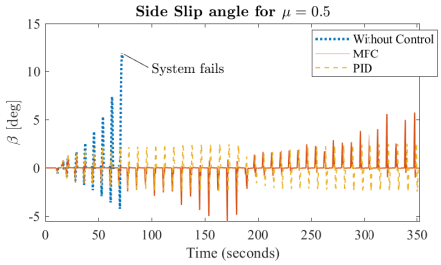


(c) Roll angle

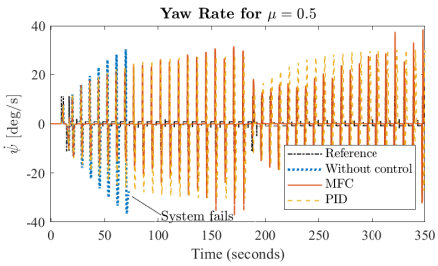


(d) Roll rate

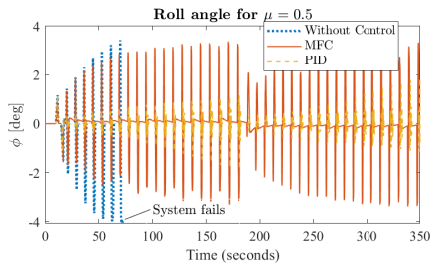
Fig. 7: Results for $\mu = 0.7$



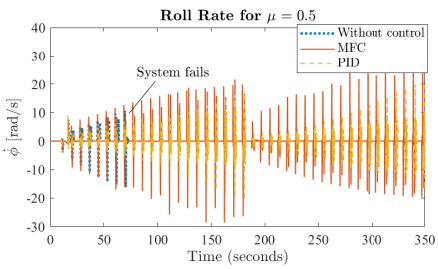
(a) Side-slip angle



(b) Yaw rate

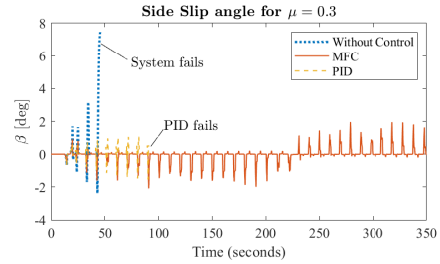


(c) Roll angle

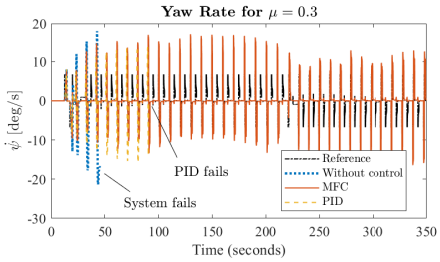


(d) Roll rate

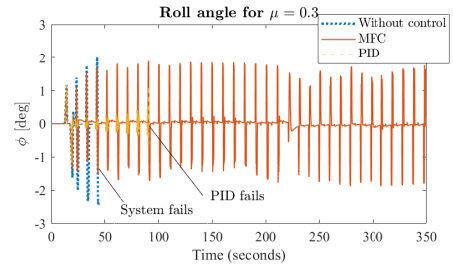
Fig. 8: Results for $\mu = 0.5$



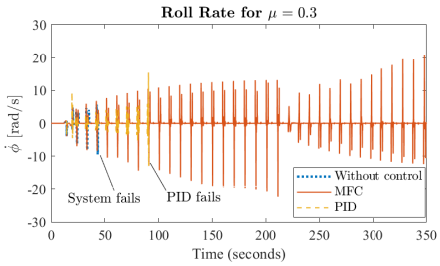
(a) Side-slip angle



(b) Yaw rate



(c) Roll angle



(d) Roll rate

Fig. 9: Results for $\mu = 0.3$

9.3 Simulation for $\mu = 0.3$

Figures 9a and 9b show the side slip angle and yaw rate respectively during the SWD maneuver with a friction coefficient of $\mu = 0.3$. It is observed that only the iPD MFC controller is able to complete the maneuver and reduce the side slip angle and yaw rate error. Both the open-loop case and the gain-scheduled LQR benchmark fail shortly after the start of the maneuver. Figures 9c and 9d show the roll angle and roll rate, where the failure can also be seen. On the other hand, the proposed MFC controller manages not only to complete the maneuver but also to maintain the roll dynamics in a desired range of values.

9.4 Statistical results across different friction conditions

Table 1 presents a statistical analysis of the vehicle's dynamic response under a Sine With Dwell (SWD) maneuver, considering the same three different road friction coefficients: $\mu = 0.7$, $\mu = 0.5$, and $\mu = 0.3$. The results are reported for the two control strategies (gain-scheduled LQR and iPD MFC) as well as the open-loop case. The cases in which stability is not guaranteed are presented in **bold numbers**.

Side-slip angle: Both controllers demonstrate a clear ability to reduce the side-slip angle. The MFC consistently achieves the lowest RMS and RMSE values, particularly at $\mu = 0.5$ and $\mu = 0.3$, where the gain-scheduled LQR benchmark shows slightly higher slip levels. At $\mu = 0.7$, MFC manages a lower RMS (0.627) compared to LQR (0.735), and at $\mu = 0.5$, this difference becomes more significant (MFC: 0.531 vs. LQR: 0.71). In the open-loop case, the side-slip angle increases substantially, peaking at 11.945° for $\mu = 0.5$, highlighting the importance of active control for maintaining lateral stability.

Yaw rate: In all conditions, the MFC achieves the lowest yaw rate RMS and RMSE values, followed by the gain-scheduled LQR benchmark. At $\mu = 0.7$, MFC maintains an RMS of 10.770 deg/s and a maximum yaw rate of 34.203 deg/s, while the uncontrolled vehicle reaches a peak of 49.98 deg/s. This pattern continues for $\mu = 0.5$, where MFC maintains better tracking and dampening of yaw motion.

Roll angle: Roll behavior also improves with both controllers. At $\mu = 0.7$, the uncontrolled vehicle experiences an extreme peak roll angle of 79.993, while MFC and the gain-scheduled LQR limit this to 5.917° and 4.161°, respectively. At $\mu = 0.5$, the LQR benchmark shows superior performance in minimizing the roll angle (RMS: 0.222), while MFC reaches a higher RMS (1.260) but still provides acceptable control.

Roll rate: Similar to the roll angle, the roll rate is substantially reduced by the control strategies. At $\mu = 0.7$, the uncontrolled vehicle reaches a peak roll rate of 115.104 deg/s, whereas the LQR and MFC controllers keep it below 42 deg/s. The same trend is observed for $\mu = 0.5$, with the MFC showing higher values than LQR at $\mu = 0.5$.

For $\mu = 0.3$, the gain-scheduled LQR benchmark and the open-loop case do not reach stability. Overall, the analysis demonstrates that both control strategies (gain-scheduled LQR and iPD MFC) significantly enhance vehicle stability and rollover mitigation during the SWD maneuver for $\mu = 0.7$ and $\mu = 0.5$, with the MFC generally offering superior performance in minimizing side-slip and yaw rate errors. The roll angle and roll rate are better under the LQR benchmark. On the other hand, the iPD MFC has better robustness of stability, as it does not fail in the case of $\mu = 0.3$. The uncontrolled vehicle (open-loop) is not robustly stable.

10 Conclusions

This study proposes a novel Unified Chassis Control (UCC) system based on Model-Free Controllers (MFCs) to simultaneously regulate vehicle lateral (side-slip angle and yaw rate) and vertical (roll angle) dynamics during abrupt maneuvers. By leveraging ultra-local models and algebraic estimators, the MFC-based UCC eliminates the need for detailed vehicle modeling, effectively addressing challenges posed by parametric uncertainties and the nonlinear nature of traditional model-based approaches. The controller was evaluated through a Sine With Dwell (SWD) maneuver in a co-simulation environment using MATLAB/Simulink® and CarSim®, under varying road friction conditions ($\mu = 0.7$, $\mu = 0.5$, and $\mu = 0.3$), and compared against a gain-scheduled LQR benchmark. Results showed that the MFC-based UCC significantly reduced side-slip angle and yaw rate errors while maintaining good roll angle stability, ensuring comprehensive lateral and vertical control. In low-friction scenarios ($\mu = 0.3$), only the MFC showed stability.

Future research should focus on automating the tuning using optimization to enhance scalability and precision, validating the approach through real-world implementations, and comparing it with alternative model-free techniques such as neural networks or fuzzy logic controllers. Additionally, extending the UCC framework to account for longitudinal dynamics or adapting it for diverse vehicle types, including electric and heavy-duty platforms, could further expand its applicability.

Table 1: Statistical analysis of simulations (lack of robustness of stability in **bold numbers**)

		GS-LQR			MFC			No Control		
Parameter	μ	RMS[1]	RMSE[2]	MAX[3]	RMS	RMSE	MAX	RMS	RMSE	MAX
Side-slip angle	0.7	0.735	0.735	2.889	0.627	0.627	3.404	**1.826**	**1.826**	**10.33**
	0.5	0.710	0.710	2.531	0.531	0.531	2.860	**2.074**	**2.074**	**11.945**
	0.3	**0.280**	**0.280**	**1.540**	0.382	0.382	2.077	**1.387**	**1.387**	**7.466**
Yaw rate	0.7	12.086	14.751	40.857	10.770	13.659	34.203	**12.730**	**15.165**	**49.980**
	0.5	9.898	11.561	30.451	8.288	10.200	26.169	**10.303**	**10.303**	**36.629**
	0.3	**4.265**	**4.265**	**15.655**	4.739	5.695	17.267	**6.203**	**6.203**	**21.354**
Roll angle	0.7	1.335	1.335	4.161	1.646	1.646	5.917	**4.642**	**4.642**	**79.993**
	0.5	0.222	0.222	1.852	1.260	1.260	4.029	**1.340**	**1.340**	**4.050**
	0.3	**0.164**	**0.164**	**1.166**	0.598	0.598	1.893	**0.804**	**0.804**	**2.452**
Roll rate	0.7	4.092	4.092	26.348	5.745	5.745	41.935	**9.844**	**9.844**	**115.104**
	0.5	2.488	2.488	20.689	4.665	4.665	37.391	**2.799**	**2.799**	**16.156**
	0.3	**1.181**	**1.181**	**15.443**	2.320	2.320	22.313	**1.528**	**1.528**	**9.482**

[1] Root Mean Square (related to signal).
[2] Root Mean Square Error (related to reference).
[3] Maximum absolute value.

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